Trainings

Preparatory Steps

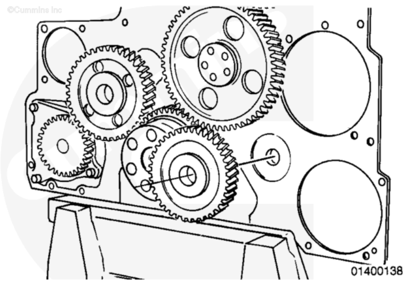
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

Remove

Note : The bolt-in idler shafts have flanges that require the shaft, the idler gear, and the thrust washer to be removed as an assembly.

Remove the capscrew from the idler gear.

Note : To remove the gear assembly, use two pry bars and pry the gear and the shaft from the block.

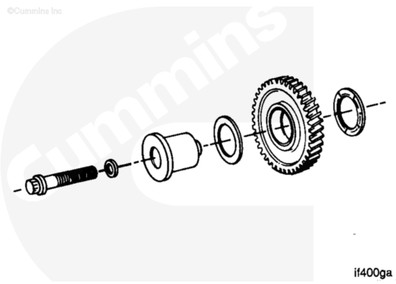


Clean and Inspect for Reuse

Use solvent. Clean the idler gear.

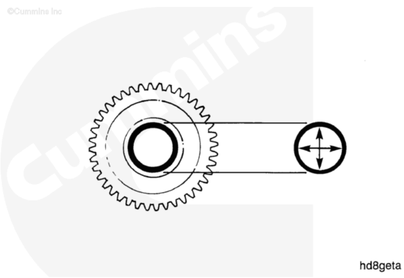
Dry with compressed air

Check the gear and parts for any damage.



Measure the inside diameter of the idler.

Idler Gear Inside Diameter		
mm		in
47.638	MIN	1.8755
47.663	MAX	1.8765

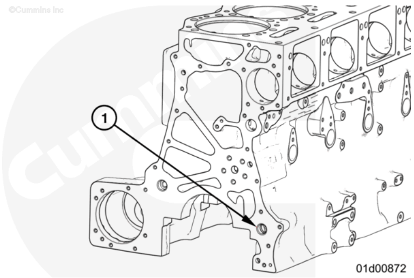


Measure the outside diameter of the idler.

Idler Gear Bushing Outside Diameter		
mm		in
47.587	MIN	1.8735
47.600	MAX	1.8740

Note : The bushing in the gear is precision bored after installation. If machining capability is not available, replace the bushing and the gear as an assembly.

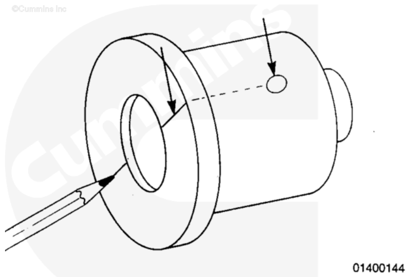
Hydraulic Pump Idler Block Bore Diameter (1)		
mm		in
25.35	MIN	0.998
25.40	MAX	1.000



Install

⚠ CAUTION ⚠

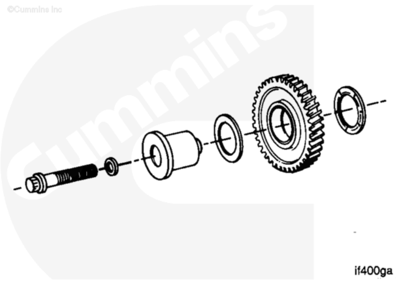
Engines that contain a hydraulic pump must have the oil holes in the hydraulic pump idler shaft as shown. Idler gear bushing failure will result if the oil holes are not aligned correctly.



Mark the flange to show oil hole orientation.

⚠ CAUTION ⚠

The grooves in the thrust washers must be turned toward the gear.

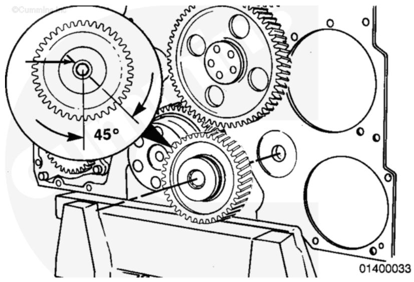


Use Lubriplate® 105 multi-purpose lubricant, or equivalent. Lubricate the gear bushing, shaft, and the thrust washer.

Use engine oil. Lubricate the capscrew. Assemble the parts as shown.

⚠ CAUTION ⚠

Engines that contain a hydraulic pump must have the oil holes in the hydraulic pump idler shaft oriented as shown. Idler gear bushing failure will result if the oil holes are not aligned correctly.



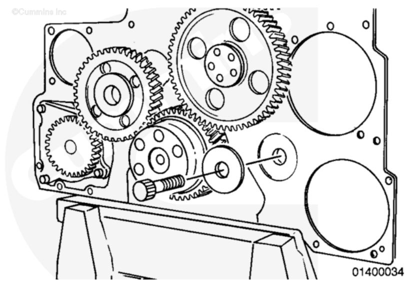
Align the oil holes in the shaft to an angle 45 degrees to the left of vertical.

Use the capscrew to pull the shaft into the bore. Tighten the capscrew.

Torque Value: 245 n•m [180 ft-lb]

⚠ CAUTION ⚠

Engines that do not have a hydraulic pump drive must have a plug installed in place of the idler shaft. Low oil pressure will result if the plug is omitted.



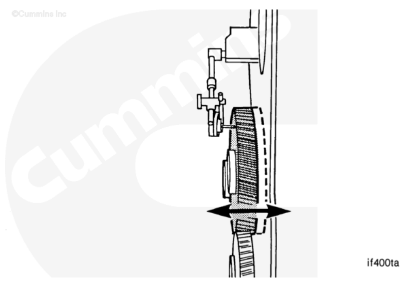
⚠ CAUTION ⚠

The capscrew that is used for the lug must not protrude beyond the block.

Use a [5/8-UNF-inch] capscrew, lock washer, and a plain washer that is larger than the block bore. Tighten the capscrews.

Torque Value: 65 n•m [50 ft-lb]

Use a dial indicator and measure the idler gear end clearance.



Hydraulic Pump Idler Gear - End Clearance

mm		in
0.10	MIN	0.004
0.36	MAX	0.014

Note : If the clearance is not within specifications, check for foreign material between the parts, or check for proper location of the thrust washers. Oversize thrust washers are available.

Finishing Steps

- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

Last Modified: 23-Feb-2022